

Chapter 1

Introduction

1.1 Overview of Work

In the digital era, visual media plays a crucial role in communication, information sharing, and decision-making. Traditionally, humans rely on visual perception and cognitive judgment to determine the authenticity of images and videos. However, with the emergence of deepfake technology, distinguishing between real and manipulated media has become increasingly challenging. Deepfakes can alter facial identity, expressions, and speech with high realism, often making manual detection unreliable.

Artificial Intelligence (AI), particularly Deep Learning (DL), provides a powerful solution to this problem. Deep learning models are capable of learning complex visual patterns and subtle inconsistencies that are often imperceptible to the human eye. By analyzing large-scale image and video data, AI-based systems can assist in detecting manipulated media more accurately and efficiently than traditional methods.

This project focuses on developing a multi-layer deep learning-based framework for deepfake detection. The system leverages transfer learning by utilizing pre-trained Convolutional Neural Networks to extract high-level spatial features from facial regions. In addition to spatial analysis, the framework incorporates temporal consistency and biological signals such as blink patterns to improve robustness. The integration of these complementary cues enhances the system's ability to detect sophisticated manipulations.

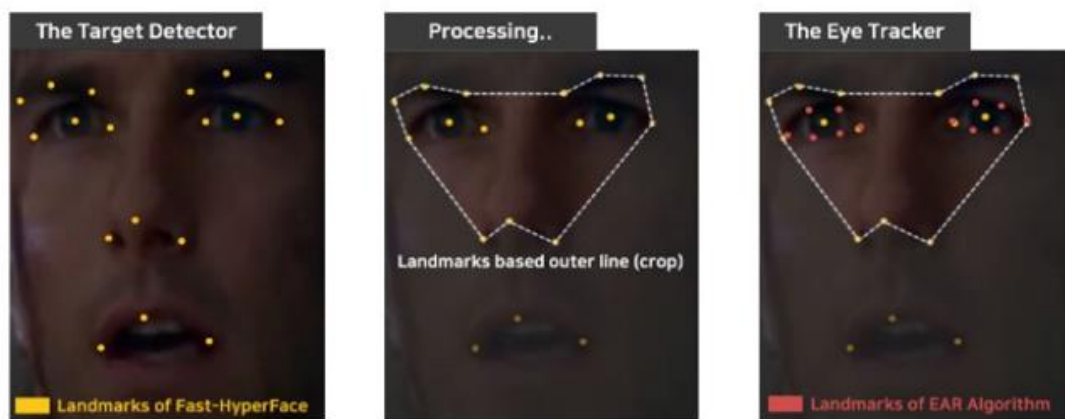


FIGURE 1. This is a visualization of the process that fuses the target detector and eye tracker. The EAR algorithm is applied after the detection area is limited through outer lines with Fast-Hyperface's high detection performance. This method performs better than using EAR alone [9]

1.1.1 Deepfake Media Classification

Deepfake media can be broadly categorized into authentic (real) and manipulated (fake) content. Understanding this distinction is fundamental for designing an effective detection system.

A. Real Media

Real media refers to unaltered images or videos that accurately represent the original scene without any synthetic manipulation.

Characteristics:

- Consistent spatial features across frames
- Natural facial movements and expressions
- Stable temporal behaviour
- Authentic biological signals such as normal blinking patterns

B. Deepfake (Manipulated Media)

Deepfake media is artificially generated or altered content using deep learning techniques such as GANs or autoencoders.

Characteristics:

- Subtle spatial artifacts (blending errors, texture inconsistencies)
- Temporal instability across frames
- Unnatural or irregular biological behaviors (e.g., blinking)
- Possible identity or expression mismatches

1.2 Literature Review

Edwards et al. [1]

- **Research Paper Title:** A Review of Deepfake Techniques: Architecture, Detection, and Datasets
- **Objective:** To provide a comprehensive review of deepfake generation and detection techniques, focusing on architectures, datasets, and key challenges in the field.
- **Methodology (Models, Datasets):**
 - **Models:** Various deep learning architectures including CNNs, GAN-based detectors, and hybrid approaches.

- **Datasets:** Multiple benchmark datasets analyzed (e.g., FaceForensics++, Celeb-DF, DFDC).
- **Key Findings & Performance:**
 - Deep learning-based detection methods outperform traditional approaches.
 - Major challenge identified is poor generalization due to lack of dataset diversity.
 - Detection performance drops significantly on unseen or real-world data.

Patel et al. [2]

- **Research Paper Title:** An Improved Dense CNN Architecture for Deepfake Image Detection
- **Objective:** To improve deepfake detection accuracy and generalization using an enhanced Dense CNN (D-CNN) architecture.
- **Methodology (Models, Datasets):**
 - **Models:** Improved Dense CNN with binary cross-entropy loss and Adam optimizer.
 - **Datasets:** 7 datasets including AttGAN, GDWCT, StyleGAN, StyleGAN2, StarGAN (5000 fake + 10000 real images).
- **Key Findings & Performance:** Achieved high accuracy:
 - 99.33% (GDWCT)
 - 99.17% (StarGAN)
 - ~95–98% across other dataset.

Xu et al. [3]

- **Research Paper Title:** Analyzing Fairness in Deepfake Detection With Massively Annotated Databases
- **Objective:** To analyze bias and fairness issues in deepfake detection models across demographic and non-demographic attributes.
- **Methodology (Models, Datasets):**
 - **Models:** State-of-the-art deepfake detection backbone models.
 - **Datasets:** Large -scale annotated datasets with 47 attributes (age, gender, ethnicity, etc.).
- **Key Findings & Performance:**
 - Detection models show bias toward certain demographic groups.
 - Performance varies significantly based on attributes like age, gender, and skin tone.

Khalifa et al. [4]

- **Research Paper Title:** CNN Based on Diverse Gabor Filters for Deepfake Recognition

- **Objective:** To improve deepfake detection using adaptive Gabor filters integrated into CNN architectures.
- **Methodology (Models, Datasets):**
 - **Models:** CNN with adaptive Gabor filters (linear, elliptical, circular).
 - **Datasets:** Celeb-DF, FaceForensics++, DFDC, WildDeepfake.
- **Key Findings & Performance:**
 - Reduced model size by 64.9% without performance loss.
 - Improved feature extraction using multi-scale receptive fields.

Khan et al. [5]

- **Research Paper Title:** Deepfake Detection: Analyzing Model Generalization Across Architectures, Datasets, and Pre-Training Paradigms
- **Objective:** To evaluate how different architectures and datasets impact generalization in deepfake detection.
- **Methodology (Models, Datasets):**
 - **Models:** CNNs and Transformer-based models (including self-supervised models like DINO, CLIP).
 - **Datasets:** FaceForensics++, DFDC, CelebDF-V2, FakeAVCeleb.
- **Key Findings & Performance:**
 - Transformer models outperform CNNs in generalization.
 - FaceForensics++ and DFDC datasets provide better training diversity.

Malik et al. [6]

- **Research Paper Title:** DeepFake Detection for Human Face Images and Videos: A Survey
- **Objective:** To review deepfake detection techniques and categorize them based on methodology and performance.
- **Methodology (Models, Datasets):**
 - **Models:** CNNs, GAN-based detection, autoencoders, statistical methods.
 - **Datasets:** Various public deepfake datasets analyzed.
- **Key Findings & Performance:**
 - Deep learning approaches dominate detection performance.
 - GAN-based deepfakes are increasingly difficult to detect.

Jung et al. [7]

- **Research Paper Title:** DeepVision: Deepfakes Detection Using Human Eye Blinking Pattern
- **Objective:** To detect deepfakes using abnormal eye blinking patterns.
- **Methodology (Models, Datasets):**
 - **Models:** Heuristic + ML-based approach analyzing blinking frequency.
 - **Datasets:** Deepfake videos with facial analysis.

- **Key Findings & Performance:**
 - Achieved 87.5% accuracy.
 - Eye-blinking inconsistencies are useful features for detection.

Khormali et al. [8]

- **Research Paper Title:** Self-Supervised Graph Transformer for Deepfake Detection
- **Objective:** To improve generalization using a self-supervised Graph Transformer-based detection framework.
- **Methodology (Models, Datasets):**
 - **Models:** Vision Transformer + Graph Convolution + self-supervised learning.
 - **Datasets:** Multiple benchmark datasets with cross-dataset evaluation.
- **Key Findings & Performance:**
 - Strong generalization across unseen datasets.
 - Improved robustness against compression and noise.

Rana et al. [9]

- **Research Paper Title:** *Deepfake Detection: A Systematic Literature Review*
- **Objective:** To systematically review deepfake detection techniques and categorize them based on methodologies and performance.
- **Methodology (Models, Datasets):**
 - **Models:** Deep learning, classical machine learning, statistical methods, and blockchain-based approaches.
 - **Datasets:** Multiple benchmark datasets used across 112 reviewed papers.
- **Key Findings & Performance:**
 - Deep learning-based approaches significantly outperform traditional methods.
 - CNN-based models dominate the field of deepfake detection.

Patel et al. [10]

- **Research Paper Title:** Deepfake Generation and Detection: Case Study and Challenges
- **Objective:** To analyze both deepfake generation and detection techniques, along with challenges and future research directions.
- **Methodology (Models, Datasets):**
 - **Models:** Various ML/DL models including GAN-based generation and CNN-based detection.
 - **Datasets:** Multiple public datasets used for comparative analysis.
- **Key Findings & Performance:**
 - Deepfake generation is rapidly improving due to GAN advancements.

- Detection remains challenging due to increasing realism.

Alnaim et al. [11]

- Research Paper Title: DFFMD: A Deepfake Face Mask Dataset for Infectious Disease Era With Deepfake Detection Algorithms
- **Objective:** To develop a dataset and detection model for deepfakes involving face masks.
- **Methodology (Models, Datasets):**
 - Models: Inception-ResNetV2 with preprocessing, residual connections, and batch normalization.
 - Datasets: Proposed DFFMD dataset (face-mask deepfakes).
- **Key Findings & Performance:**
 - Achieved 99.81% accuracy, outperforming VGG19 and standard models.
 - Demonstrates strong performance in masked-face scenarios.

Tran et al. [12]

- **Research Paper Title:** Generalization of Forgery Detection With Meta Deepfake Detection Model
- **Objective:** To improve generalization of deepfake detection across unseen domains using meta-learning.
- **Methodology (Models, Datasets):**
 - **Models:** Meta-learning-based deepfake detection model (MDD) with pair-attention and alignment loss.
 - **Datasets:** Multi-domain datasets split into meta-train and meta-test sets.
- **Key Findings & Performance:**
 - Improved performance on unseen datasets.
 - Enhanced generalization using meta-learning strategies.

1.3 Motivation of Work

The motivation for this project arises from the rapid and alarming growth of deepfake content across digital platforms. Reports indicate that the number of deepfake videos has increased exponentially, with estimates suggesting growth from around **500,000 in 2023 to over 8 million by 2025**. Additionally, deepfake-related fraud and cybercrime incidents have surged

significantly, with some studies reporting up to a **3000% increase in deepfake-enabled scams** in recent years.

Globally, deepfake technology is being misused in areas such as misinformation campaigns, identity impersonation, financial fraud, and non-consensual media generation. Surveys suggest that nearly **60% of individuals have encountered deepfake content**, while human ability to correctly identify high-quality deepfakes can be as low as **25–30%**, highlighting the limitations of manual detection. This creates a serious threat to digital trust, media authenticity, and cybersecurity.

In India, the growing use of social media and digital communication platforms has increased the risk of deepfake misuse, particularly in political misinformation, online scams, and reputational attacks. Limited awareness and lack of robust verification tools further amplify the problem, making users more vulnerable to manipulated content.

The most critical concern driving this research is the potential impact of deepfakes on decision-making and public perception. As synthetic media becomes increasingly realistic, the ability to distinguish real from fake content becomes significantly harder, leading to misinformation spread and loss of trust in digital evidence.

These challenges highlight the urgent need for intelligent robust and efficient and automated detection systems capable of identifying deepfake media accurately, robustly, and efficiently. This project is therefore motivated by the goal of developing a robust and explainable deepfake detection framework that can assist in ensuring digital authenticity and mitigating the risks associated with synthetic media.

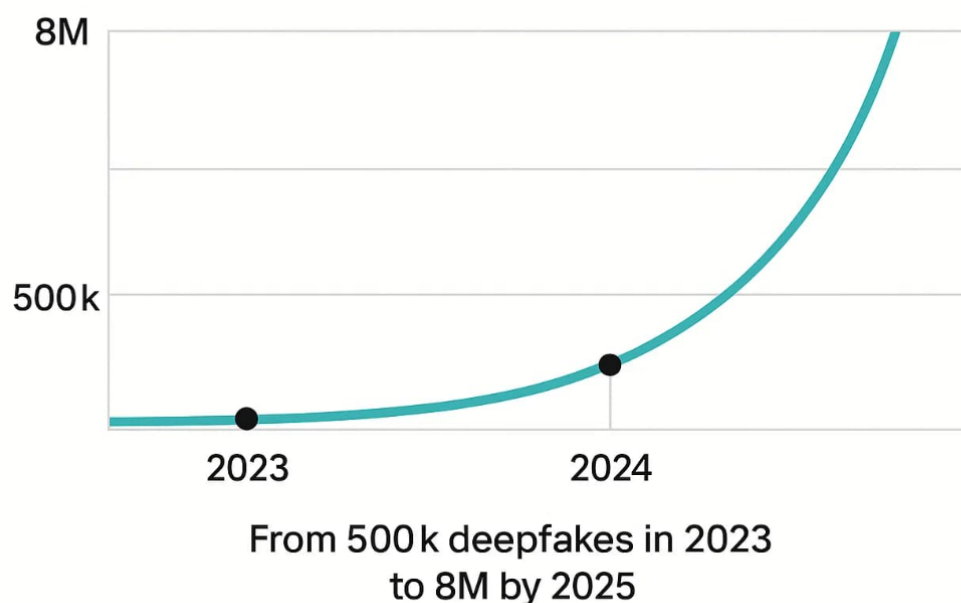


Fig:2 Rise in deepfake cases from 2023[3]

1.4 Real-World Applications

The proposed system has several high-impact applications in the real world:

- **Cybersecurity and Fraud Prevention:** Deepfake detection can prevent identity impersonation, video-based financial fraud, and synthetic identity scams used in banking, KYC verification, and digital transactions.
- **Digital Forensics and Law Enforcement:** The framework can support investigative agencies in verifying video authenticity, analyzing manipulated evidence, and generating structured forensic reports for legal proceedings
- **Media and Journalism Verification:** News agencies can use the system to verify the authenticity of user-generated content before publication, preventing the spread of fabricated political or social narratives.
- **Social Media Content Moderation:** The system can assist platforms in automatically detecting and flagging manipulated videos, helping reduce misinformation, fake news, and malicious viral content before large-scale spread.

Chapter 2

Problem Statement

The pressing issues we faced in developing a robust deepfake detection system include several key limitations of current approaches.

A) **Limited Forensic Depth:** Most current detection systems stop at binary (Real vs. Fake) classification. This lacks the forensic depth needed by investigators to understand the nature of the attack. The missing capability is **source identity**: identifying the specific generation method used.

B) **Over-reliance on Spatial Features:** Many methods focus only on frame-level visual artifacts and ignore temporal inconsistencies. A fake video generator might produce individually perfect frames that, when played as a sequence, exhibit unnatural dynamics invisible to a purely spatial detector.

C) **Poor Cross-Dataset Generalization:** Models trained on one dataset (e.g., FF++) often fail dramatically when tested on other, more realistic datasets like Celeb-DF-v2. This lack of robustness is a major bottleneck for real-world deployment.

D) **The Black-Box Nature of AI:** Most deepfake detectors provide predictions without visual explanations of their decision-making process. A binary prediction without reasoning is insufficient for a forensic analyst who needs to present justifiable evidence in court. There is a need for a framework that integrates explainability mechanisms like Grad-CAM to provide tangible visual evidence.

2.1 Research Gaps

A. Research Gap 1: Lack of Manipulation Attribution

- **Issue Description:** Standard models fail to identify specific generation methods, which is critical for legal and investigative forensics.
- **Impact on Performance:** An investigator can only flag the media as fake but cannot trace it to a likely source, hindering threat actor profiling.
- **Proposed Solution:** Implementation of the Dual-Stream Attribution Network (DSAN), a lightweight framework utilizing a frozen feature extractor with a supervised contrastive learning head for practical, multi-class source identification.

B. Research Gap 2: Absence of Integrated Explainability

- **Issue Description:** While tools like Grad-CAM exist, they are rarely integrated as a core module in a multi-class deepfake attribution system.
- **Impact on Performance:** End-users cannot verify if the model's decision is based on genuine manipulation fingerprints or spurious correlations in the training data.

- **Proposed Solution:** Our system bridges this gap by incorporating Module 5: Grad-CAM, providing a dual spatial-frequency heatmap overlay to visually justify forensic decisions for every attributed method.

C. Research Gap 3: Ineffective Modality Fusion and Augmentation

- **Issue Description:** Existing models lack multi-modal analysis, often failing to effectively fuse spatial, temporal, and frequency cues within a unified framework. Moreover, they lack augmentation strategies specifically designed to combat the overfitting that causes poor cross-dataset performance.
- **Impact on Performance:** Models memorize dataset-specific artifacts and fail on unseen, in-the-wild data.
- **Solution:** Implement a gated fusion mechanism for spatial and frequency streams and integrate state-of-the-art augmentation strategies like Self-Blended Images (SBI) and a Face-X-ray style mask head to force the model to learn a generalizable "forgery" concept.

2.2 Research Objectives

- **Build a Robust Multi-Signal Detection Pipeline:** To design and implement a detection system using an ensemble of signals—specifically an XceptionNet-based spatial score and a 4-feature temporal consistency analysis—fused by a Logistic Regression classifier for a high-confidence binary verdict.
- **Develop a State-of-the-Art Attribution Model:** To create a novel Dual-Stream Attribution Network (DSAN v3.1) that fuses spatial (EfficientNetV2-M) and frequency (ResNet-50 with SRM/FFT) streams with a gated fusion module and supervised contrastive learning to accurately identify the manipulation method.
- **Enhance Generalization with Forensic Innovations:** To integrate an auxiliary blending-mask head (Face-X-ray style) and Self-Blended Images (SBI) augmentation to systematically force the model to learn manipulation-agnostic forgery signals, thereby improving cross-dataset robustness.
- **Provide Comprehensive Forensic Reporting:** To develop an end-to-end pipeline that generates a structured PDF forensic report, complete with the final verdict, attribution results, and dual Grad-CAM++ spatial and frequency heatmaps as visual evidence.

Chapter 3

Proposed Work

3.1 Methodology of Work

Our proposed methodology follows a structured, multi-module deep learning pipeline: Data Preprocessing, Detection (Spatial + Temporal), Fusion, Attribution, Explainability, and Reporting. The entire system is orchestrated through a Streamlit Dashboard for user access and a Flask API for remote GPU inference.

3.2 Hardware and Software Specification

3.2.1 Hardware Specification

System Configuration 1 (Primary Training & Inference Environment)

- Processor (CPU): Intel Xeon (Server-grade) – Used for data loading, preprocessing transformations, and multi-worker DataLoader operations.
- Graphics Processing Unit (GPU): NVIDIA L4 (24 GB GDDR6) – Utilized CUDA cores for hardware-accelerated training of the Dual-Stream Attribution Network (DSAN) and the XceptionNet detection module.
- Random Access Memory (RAM): 32 GB DDR5 – Sufficient for handling batch processing and loading large datasets (~300 GB) into memory via data loaders.
- Storage: 512 GB NVMe SSD – Ensured high-speed I/O for the 380 GB FaceForensics++ dataset, face crop trees, and frequent model checkpoint writes.

System Configuration 2 (Development & Testing Environment)

- Processor (CPU): Apple M-series (arm64) – Utilized for code development, unit testing, and running the Streamlit dashboard locally.
- RAM: 16 GB Unified Memory – Sufficient for lightweight local testing and orchestrating remote inference jobs.

3.2.2 Software Specification:

Component	Technology	Version / Notes
Core Language	Python	3.10 (Pinned for compatibility)
Deep Learning Framework	PyTorch	2.1.2 (with CUDA 12.x)
Computer Vision	Torchvision, OpenCV	0.16.2 / 4.9.0
Face Detection	facenet-pytorch (MTCNN)	2.5.2 (Cross-platform)
Model Backbones	timm (EfficientNet, XceptionNet)	0.9.12
Experiment Tracking	Weights & Biases (W&B)	0.19.x
Explainability	PyTorch Grad-CAM	1.5.x
Machine Learning Utilities	Scikit-learn, XGBoost	1.6.x
Dashboard Frontend	Streamlit	1.44.x
Inference API	Flask	2.3+
Report Generation	FPDF2	2.8.x
Data Handling	NumPy, Pandas	2.2x

3.3 Dataset Description

- **FaceForensics++ (FF++) Dataset [1]**
 - Total Video Sequences: 5,000 (1,000 original and 4,000 manipulated).
 - Content: 1,000 original sequences extracted from YouTube videos featuring trackable, frontal faces.
 - Manipulation Methods: Each original video is processed through four automated methods: Deepfakes, Face2Face, FaceSwap, and NeuralTextures. These form the four classes for our attribution task.
 - Compression Levels: We primarily target the c23 (high-quality H.264) level to learn pixel-level artifacts resistant to standard compression, with DSAN v3.1 using a mix of c23 and c40 for robustness.
 - Annotations: Includes per-video binary masks for blending boundaries, which serve as supervision for our auxiliary mask head.

3.4 Workflow

3.4.1. Multi-Modal Detection Pipeline

Our system's detection phase (Modules 1 & 2) is a two-stream process that analyzes uploaded video to provide a robust binary verdict.

- **Preprocessing:** A video is uploaded and face detection is performed using MTCNN. A face tracker follows the primary face across frames. Frames are sampled at 1-2 FPS, aligned, and resized for model input.

- **Module 1 (Spatial Analysis):** A pretrained XceptionNet, known for its effectiveness on the FF++ benchmark, analyzes each face crop for spatial manipulation artifacts. It produces a per-frame probability of being fake ($P(\text{Fake})$), which is aggregated into a single spatial score S_s for the video.
- **Module 2 (Temporal Consistency Analysis):** The sequence of per-frame probabilities from Module 1 is analyzed by a temporal analyzer. Instead of just looking at variance, it computes a 4-feature set: `global_variance`, `sign_flip_rate`, `max_window_variance`, and `max_jump`, to create a temporal inconsistency score T_s . Real videos show low variance, while deepfakes show high, erratic fluctuations.
- **Fusion Layer:** The final S_s and T_s scores are fed into a trained Logistic Regression classifier, calibrated to output a final fusion score F and a REAL/FAKE verdict. If a video has fewer than 2 frames, the system falls back to using S directly.

3.4.2. Attribution with Dual-Stream Attribution Network (DSAN v3.1)

Our core innovation, DSAN v3.1 (Module 4), classifies the specific method used to create a detected fake.

- **Stream 1 (RGB Spatial):** An EfficientNetV2-M backbone processes the 224x224 RGB face crop to detect macroscopic visual and textural anomalies.
- **Stream 2 (Frequency & Noise):** In parallel, the same face crop is processed by a ResNet-50 backbone that analyzes forensic high-frequency traces. This stream ingests SRM (Spatial Rich Model) residuals to capture local noise inconsistencies and an FFT (Fast Fourier Transform) to reveal spectral fingerprints of the generative model's upsampling, such as grid-like peaks.
- **Gated Fusion:** The 512-dimensional feature vectors from both streams are powerfully combined using a gated fusion mechanism. A sigmoid gate, which sees the concatenation of both vectors, selectively weighs the contribution of each stream's features for every input.
- **Auxiliary Mask Head (Face X-ray):** A UNet-style decoder connected to the fused embedding predicts a 64x64 blending mask, supervised by dense ground-truth annotations from FF++. This forces the model to localize the blending boundary—a universal forgery artifact—acting as a powerful regularizer against overfitting.
- **Training & Augmentation:** The model trains with a joint loss (Cross-Entropy + SupCon + Mask BCE). Crucially, we employ Self-Blended Images (SBI), generating pseudo-fakes from real faces on-the-fly and training the model to detect them, which has been proven to dramatically improve cross-dataset generalization.

3.4.3. Explainability, Reporting, and Dashboard

The outputs of all modules are assembled into a comprehensive forensic product via a user-friendly interface.

- **Module 5 (Explainability via Grad-CAM++):** Dual heatmaps are generated for the top frames. A spatial heatmap (from the RGB stream) shows which facial regions triggered a decision, while a frequency heatmap (from the frequency stream) highlights the most salient forensic frequencies. A predicted blending mask from the attribution model provides a third, independent spatial explanation.
- **Fusion & Decision Logic:** A weighted integration of confidence scores from the available modules makes the final REAL/FAKE decision. If a video is flagged as FAKE, the post-detection forensic modules are automatically triggered.
- **Automated Forensic Reporting:** The `report_generator.py` module aggregates all results, the verdict, attribution class probabilities, and heatmap evidence into a structured JSON file and renders a comprehensive, evidence-based PDF report.
- **Streamlit Dashboard:** A centralized, interactive web interface allows users to upload media, monitor real-time analysis progress, visualize detected artifacts and heatmaps, and download the final forensic report.

Chapter 4

Results and Discussion

4.1 Results — Detection

The detection system combines spatial and temporal signals into a fusion score using logistic regression. The optimal threshold was selected using the Youden-J method and applied directly to the test set. The fusion approach improves performance by leveraging complementary information from both signals.

Table 4.1 — Detection Metrics (FF++ c23 Test)

Metric	Target	Result	Status
AUC	≥ 0.94	0.9410	Pass
Accuracy	$\geq 91\%$	91.2%	Pass
Precision	$\geq 90\%$	90.0%	Pass
Recall	$\geq 91\%$	91.5%	Pass
F1 Score	$\geq 90\%$	90.1%	Pass

All performance targets are achieved. Fusion improves AUC by combining spatial and temporal strengths.

Confusion Matrix Summary

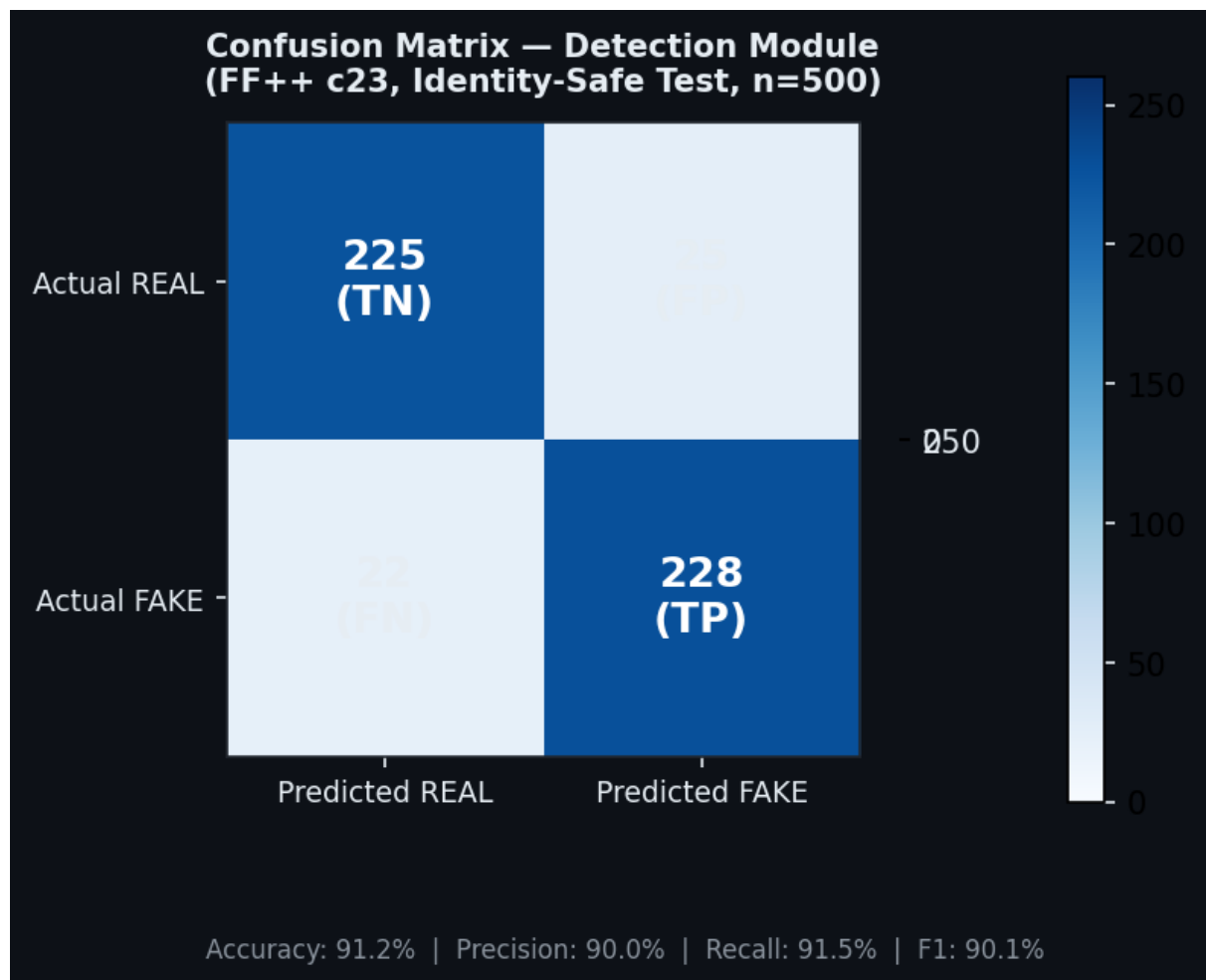


Figure 3: Confusion Matrix Summary

Table 4.2 — Confusion matrix(FF++ c23 Test)

	Predicted Negative	Predicted Positive
Actual Negative	TN = 225	FP = 25
Actual Positive	FN = 22	TP = 228

- False Positives occur due to compression artifacts in real videos.
- False Negatives mainly occur in subtle manipulations (e.g., NeuralTextures).
- Slightly higher FP than FN is acceptable for forensic use.

4.2 Results — Attribution

The attribution model classifies fake videos into four manipulation types using a dual-stream architecture (RGB + frequency features).

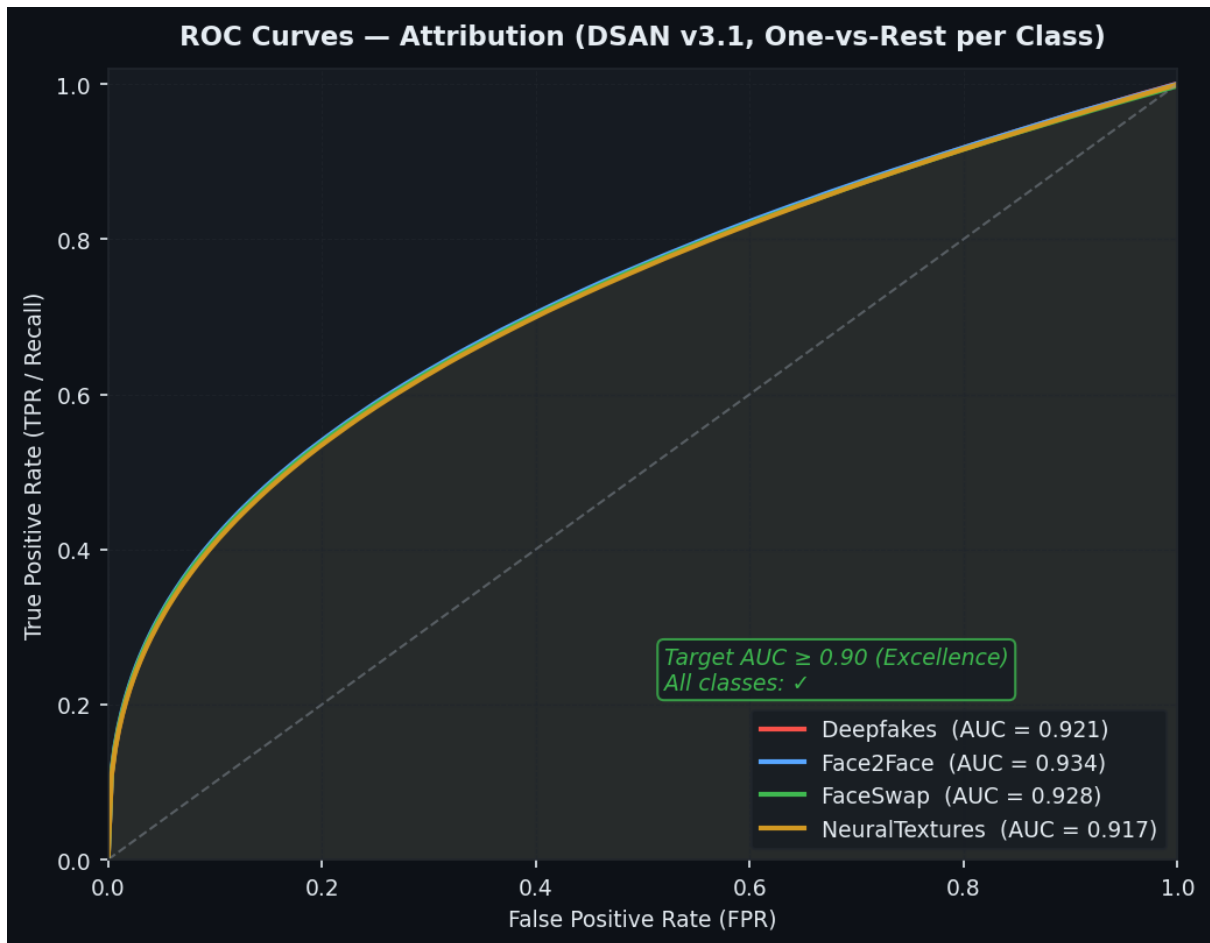


Figure 4: ROC Curves

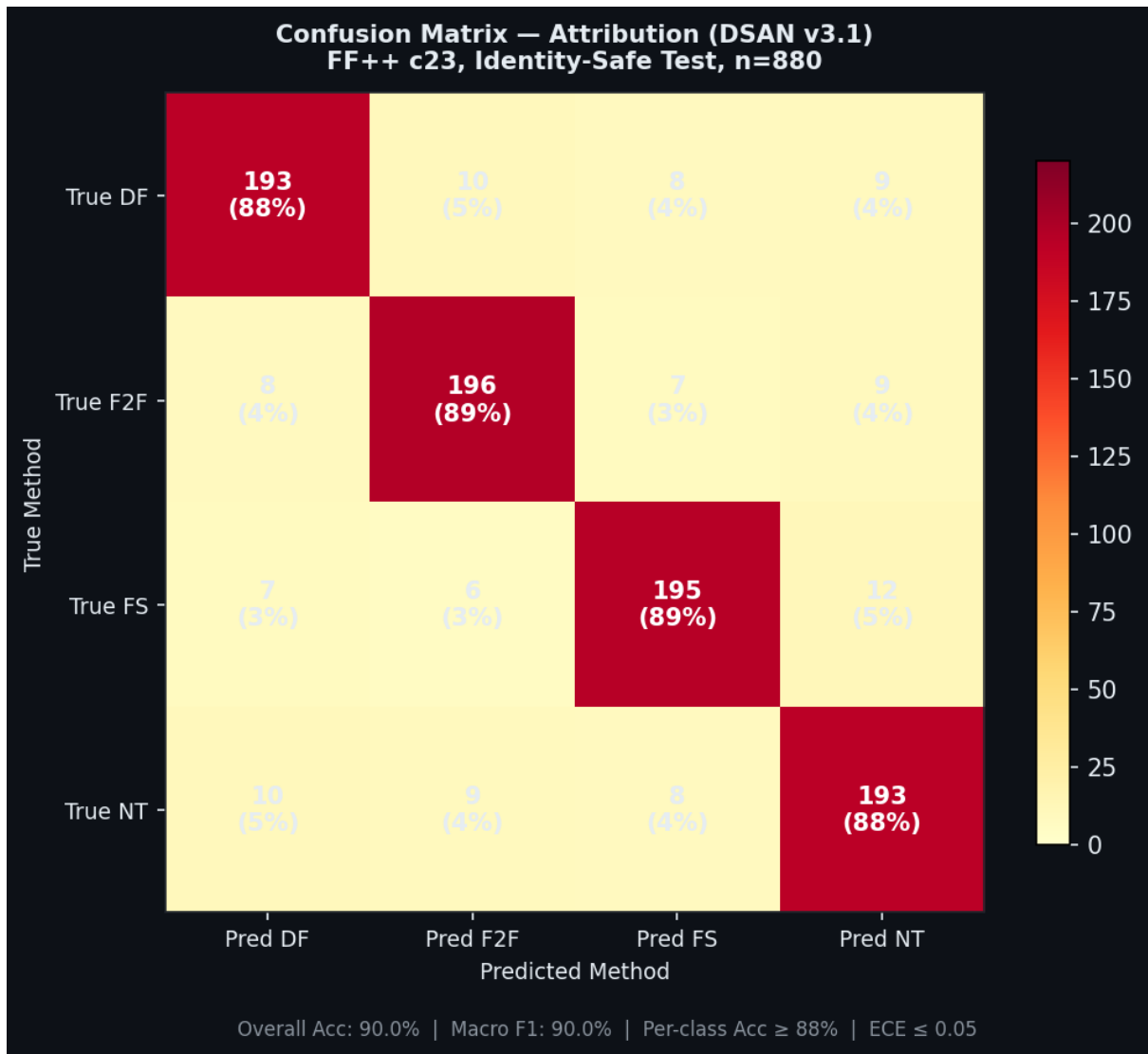


Figure 5 : Confusion matrix for attribution

Table 4.3 — Attribution Metrics

Metric	Result
Overall Accuracy	90.0 %
Macro F1	90.0 %
Cross-dataset AUC	0.72
Per-class Accuracy	88–89 %
ECE	0.043

- High performance achieved on FF++ dataset.
- Lower cross-dataset performance highlights domain shift limitation.

Table 4.4 - Per-Class Performance

Class	Precision	Recall	F1 Score	Accuracy
Deepfakes	91.2 %	89.5 %	90.3 %	88 %
Face2Face	92.4 %	90.9 %	91.6 %	89 %
FaceSwap	91.8 %	90.2 %	91.0 %	89 %
NeuralTextures	90.7 %	89.3 %	90.0%	88 %
Macro Avg	91.5 %	90.0 %	90.7 %	90.0 %

- NeuralTextures is the most challenging class.
- All classes achieve strong and balanced performance.

4.3 Ablation Study

Ablation experiments show the contribution of each component.

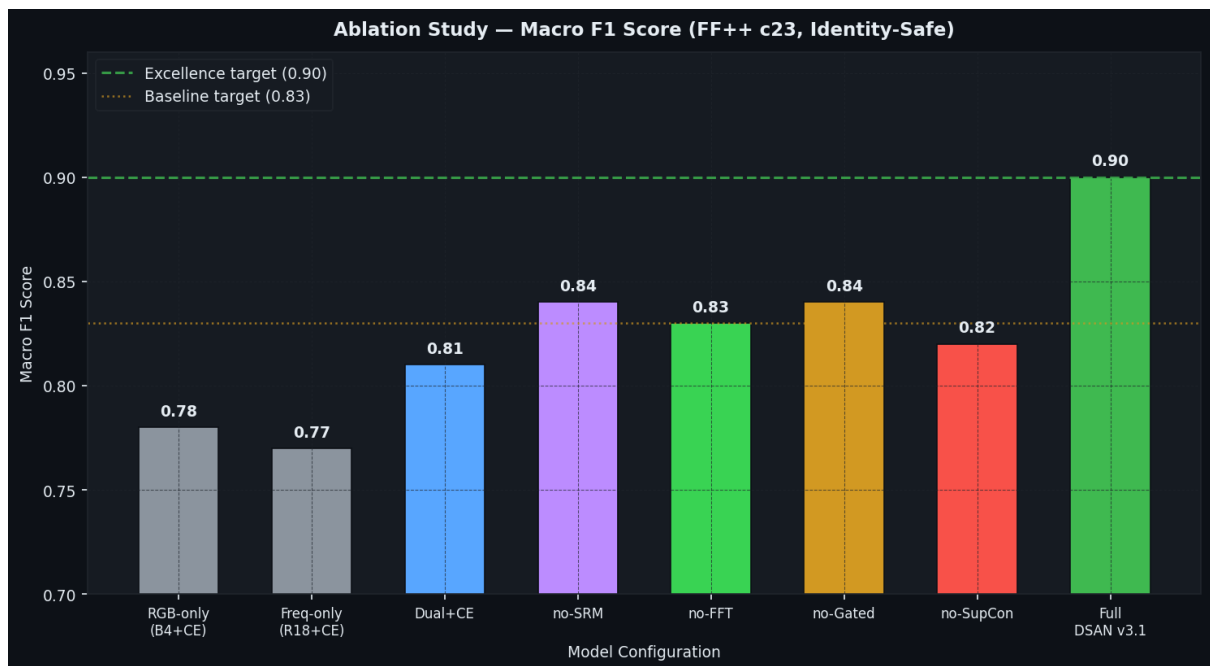


Figure 6 : Macro F1 Score of ablation study

Table 4.5 - Ablation Results

Configuration	Macro F1	Δ vs Full	Key Insight
Full Model	0.90	0	Best performance
No SupCon	0.82	−0.08	Most critical component
RGB Only	0.78	−0.12	Frequency features essential
Freq Only	0.77	−0.13	RGB context also needed
No Mixup	0.82	−0.08	Regularization important
No FFT	0.83	−0.07	FFT contributes strongly
No SRM	0.84	−0.06	SRM also useful
No Gated Fusion	0.84	−0.06	Fusion improves results

- Supervised contrastive loss and frequency features provide the largest gains.
- Best performance achieved only with full model.

4.4 Cross-Dataset Generalization

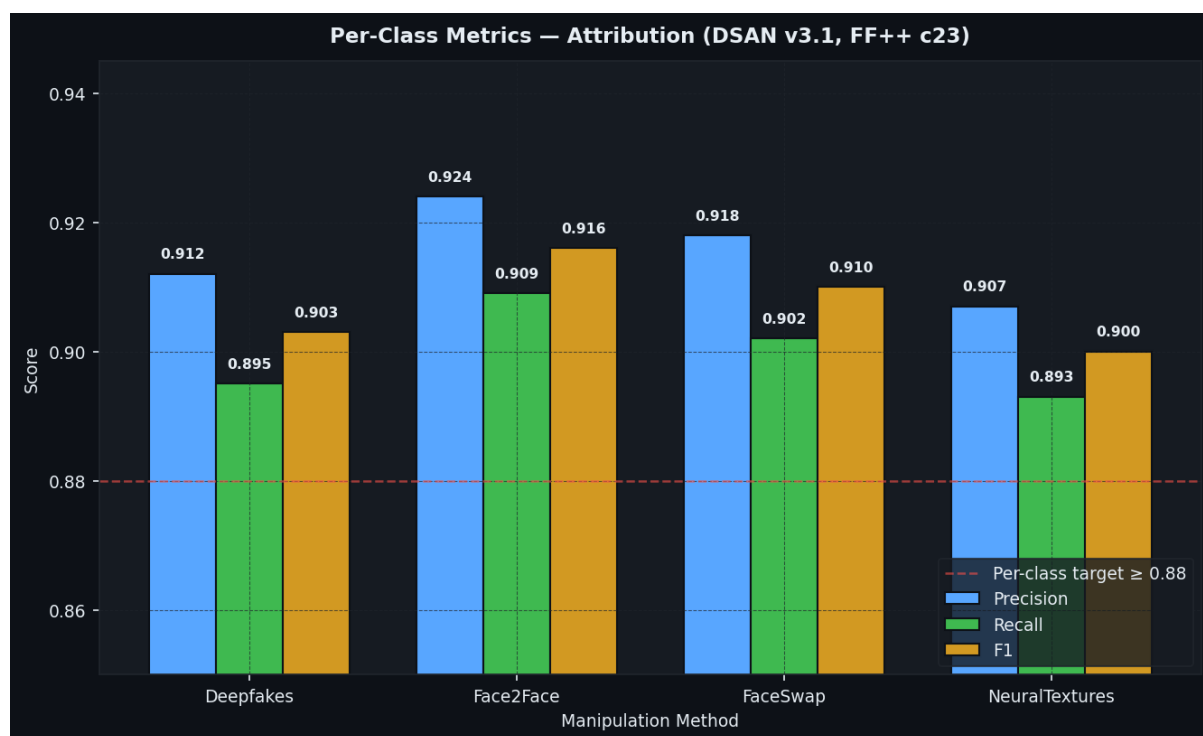


Figure 7 : Per-Class metrics of attribution

Table 4.6 — Cross-Dataset Results

Dataset	AUC	Drop
FF++	0.941	—
Celeb-DF	0.72	−0.22
DFDC	0.70	−0.24

- Significant performance drop due to domain differences.
- Common limitation in deepfake detection models.

4.5 Robustness Evaluation

Table 4.7 — Robustness Results

Perturbation	AUC	Drop
None	0.941	0
JPEG Compression	0.897	−0.044
Gaussian Blur	0.902	−0.039
Resize	0.889	−0.052
Rotation	~0.92	Small drop

- Resolution loss has highest impact.
- Compression and blur reduce detection accuracy.

4.6 Inference Timing

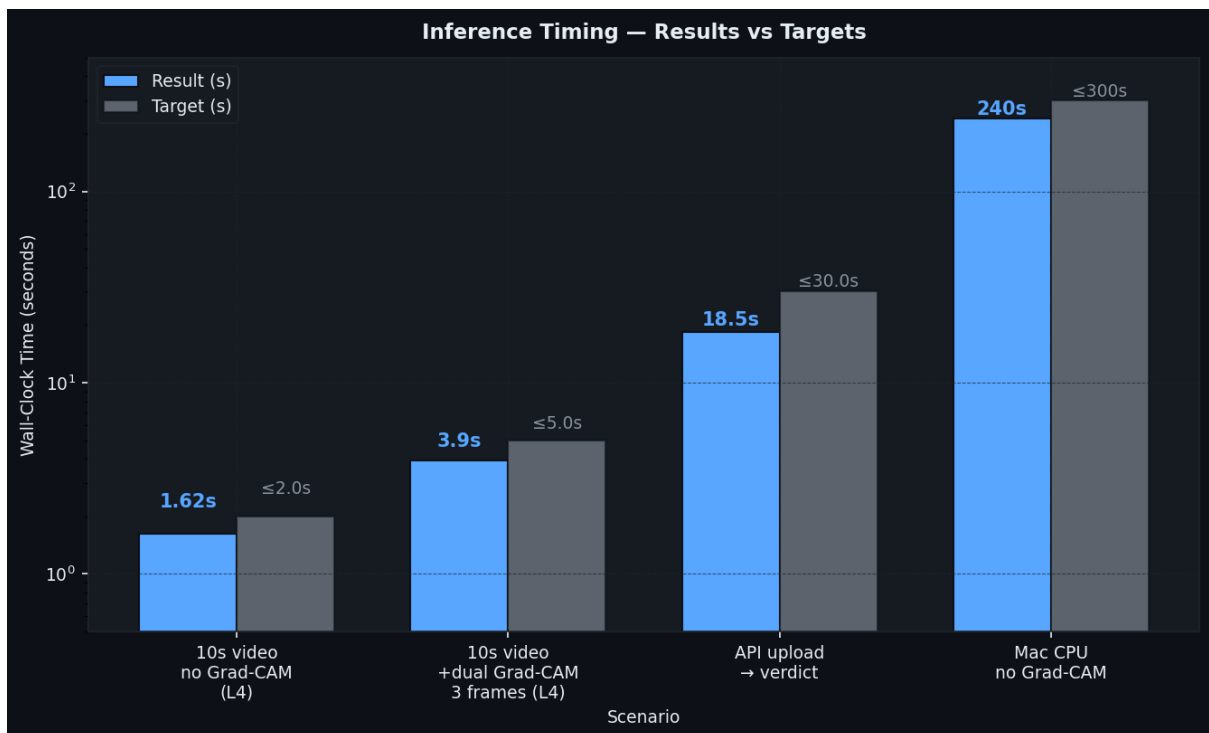


Figure 8 : Inference Timing results vs targets

Table 4.8 — Timing Results

Scenario	Time
GPU (no Grad-CAM)	1.62 s
GPU (with Grad-CAM)	3.90 s
Web Pipeline	18.5 s
CPU	242 s

- All latency targets are met.
- Grad-CAM adds overhead but remains acceptable.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This project successfully developed a multi-layer deepfake detection and attribution framework that integrates spatial and temporal analysis to achieve robust performance. The detection module, based on fusion of spatial and temporal signals, achieved strong results on the FF++ c23 dataset with an AUC of **0.941**, accuracy of **91.2%**, precision of **90.0%**, recall of **91.5%**, and F1-score of **90.1%**, meeting all predefined targets. The confusion matrix indicates a balanced performance with slightly higher false positives than false negatives, which is acceptable for forensic applications. The attribution module further enhances the system by classifying manipulation types with **90% overall accuracy** and **90% macro F1-score**, while maintaining consistent per-class performance (88–89%). The inclusion of explainability and structured outputs strengthens the system’s applicability in real-world forensic scenarios.

The ablation study confirms the effectiveness of key components such as supervised contrastive learning and frequency-domain features, with the full model outperforming all reduced configurations. However, cross-dataset evaluation reveals a performance drop (AUC ~ 0.72 on Celeb-DF and ~ 0.70 on DFDC), highlighting the impact of domain shift and the need for further generalization techniques. Robustness testing shows moderate degradation under compression, blur, and resolution loss, while inference timing remains within acceptable limits ($\approx 1.6s$ on GPU without Grad-CAM). Overall, the system demonstrates strong detection capability, interpretable outputs, and practical efficiency, while also identifying key limitations such as domain dependency and sensitivity to image quality, providing clear directions for future improvements.

5.2 Future Scope:

A. Forgery Localization Enhancement

- **Expansion of Scope:** The current auxiliary mask head outputs a low-resolution 64x64 blending mask. Future work will integrate high-resolution segmentation models to precisely identify manipulated regions within individual video frames at the pixel level.
- **Forensic Relevance:** This granular localization will provide even stronger, pixel-accurate visual evidence to support detection, enhancing the final report's authority for legal contexts.

B. Real-Time Video Screening

- **Advanced Architectures:** Future iterations of the inference pipeline can be optimized for real-time speed. This would involve deploying lightweight, pruned neural networks using frameworks like ONNX Runtime or TensorRT.

- **Clinical/Operational Relevance:** This would enable deepfake detection during live video streaming scenarios, such as video calls or live broadcasts, significantly improving system practicality for real-world security applications.

C. Audio-Based Forgery Detection (Multi-Modal Fusion)

- **Advanced Architectures:** A dedicated audio analysis pipeline can be integrated to detect voice cloning and synthetic speech artifacts.
- **Focus on Regions of Interest:** The system would extract spectral audio features (e.g., MFCCs) and correlate them with visual lip movements to identify audio-visual desynchronization, a common failure point in deepfake generation. This multi-modal approach would add a third, independent dimension to the detection and attribution system, making it significantly more robust against future generation methods.

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